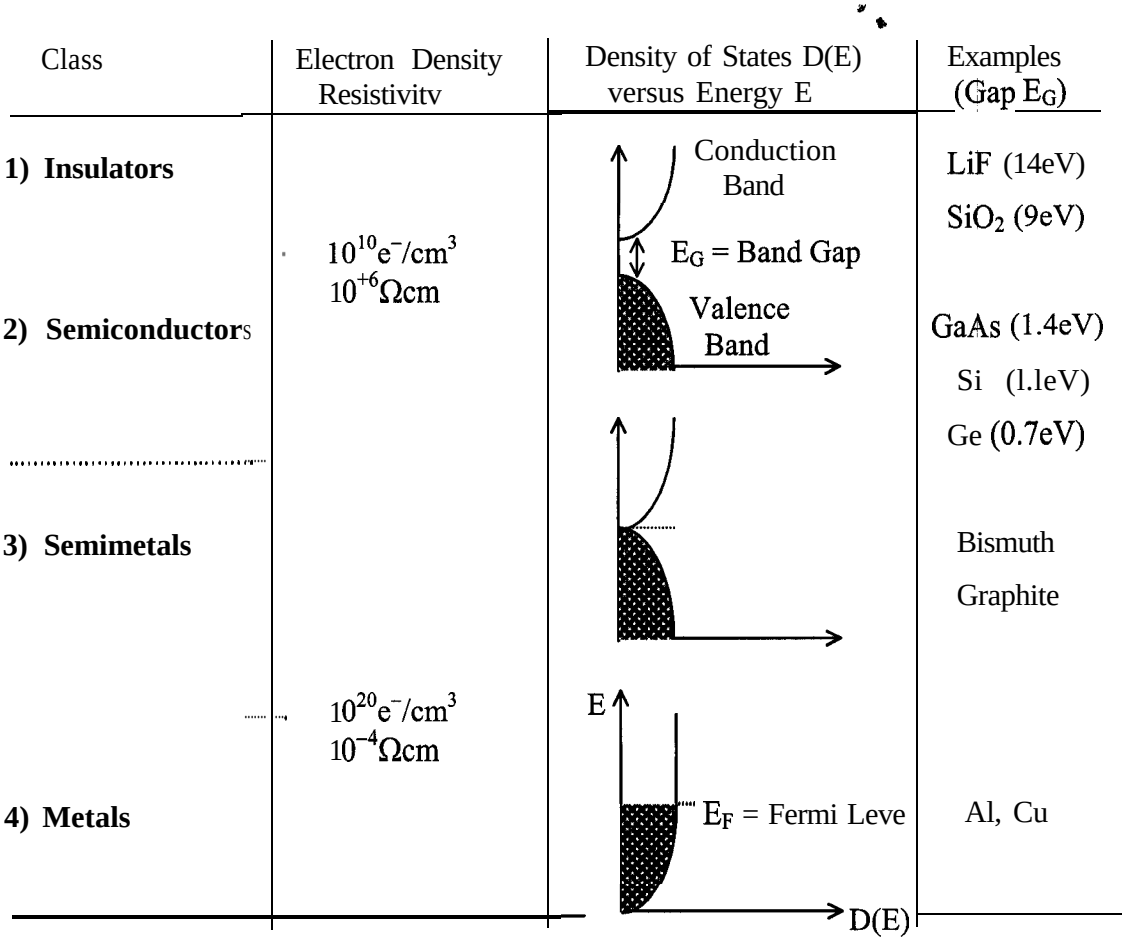
Classification of Solids

Classification by **conductivity,** which is related to the band structure:

(Filled bands are shown dark; D(E) = Density of states)



Energy scales: Band width $\approx 10^1$eV, Band gap $\approx 10^0$eV, kT $\approx$ 25meV (room temperature)

Metals: Resistance **increases** with increasing temperature.

Thermal lattice vibrations (phonons) scatter electrons.

Semiconductors / Insulators: Resistance **decreases** with increasing temperature.

Thermal excitation of electron/hole pairs (carriers).

Semiconductors

Conductivity

$\mathbf{J} = \sigma \cdot \mathbf{E}$ defines **conductivity** σ (**J** = current density, **E** = electric field)

$\mathbf{v} = \pm\mu \cdot \mathbf{E}$ defines **mobility** μ ($\mu>0$ by definition, v = drift velocity)

$\mathbf{J} = n \cdot q \cdot \mathbf{v}$ n = carrier density (n_h ,n_e) q = carrier charge (+e,–e)

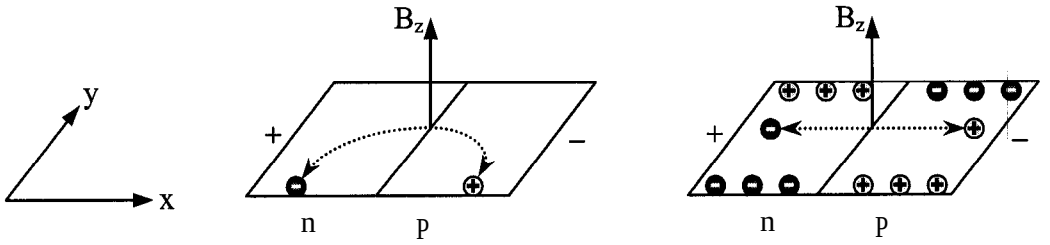
$\mathbf{v} = (q\tau / m^*) \cdot \mathbf{E}$ τ = scattering time m^* = effective mass

$\sigma = (n\, e^2\, \tau) / m^*$

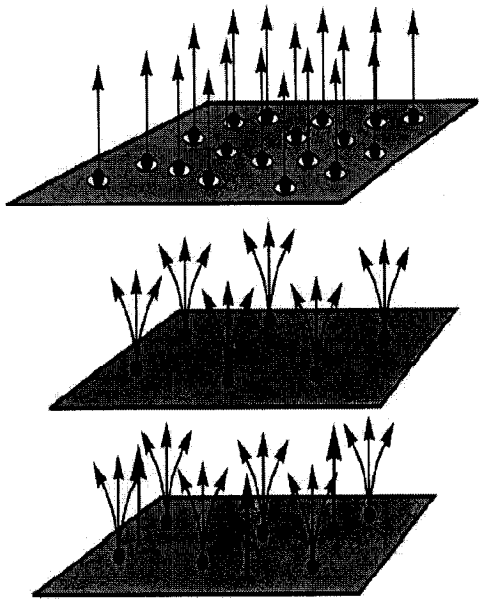
$\mu = (e\, \tau) / m^*$

Hall Effect

$\frac{E_y}{J_x B_z} = \frac{1}{\rho\, q} = R_H$ **Hall voltage** $V_H = y \cdot E_y$ provides carrier density ρ and sign of q.



Quantum Hall Effect



Integer Quantum Hall Effect:
Electrons (dots) move in circular orbits and populate discrete Landau levels (magnetism notes p. 6). For a filling v=l of the lowest Landau level each electron orbits around one vortex (= magnetic flux quantum h/e, arrows).

Fractional Quantum Hall Effect:
For a filling $\nu=1/3$ each electron is associated with three vortices.

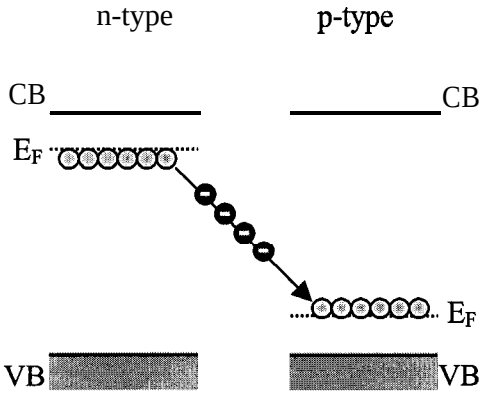
Fractional charge of e/3 for a hole (= single vortex without its share of 1/3 electron). Such extra vortices appear when B is higher than required for a filling $\nu=1/3$.

Semiconductor Junctions

Start out with two neutral pieces. Put them into contact and let electrons flow to lower states until the two Fermi are equilibrated. Thereby, neutral dopants become ionized ("space charge") and an electrostatic potential builds up. E_F is shown for T=O.

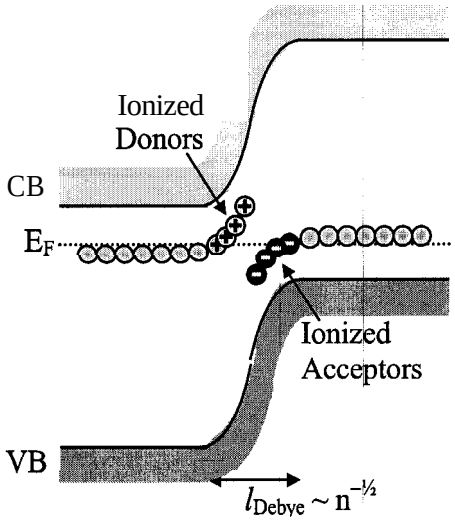
pn -Junction

Separate p- and n-type semiconductors:



Electrons flow downhill from n to p until the Fermi levels E_F line up.

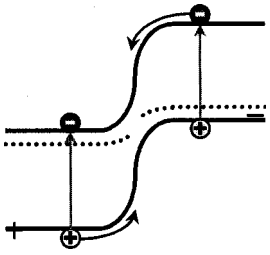
After forming the pn -junction:



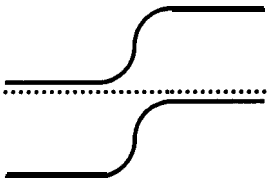
Depletion:
E_F mid-gap

Rectifying Diode

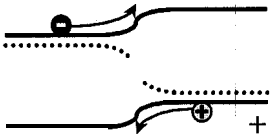
Reverse bias



Zero bias

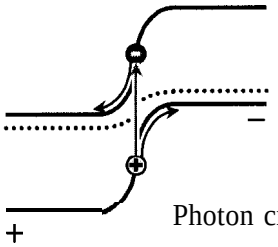


Forward bias



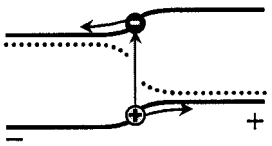
Photodiode

Reverse bias



Solar Cell

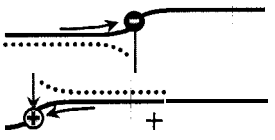
Forward bias



Photon creates e^-h^+ pair

LED (Light Emitting Diode)

Forward bias

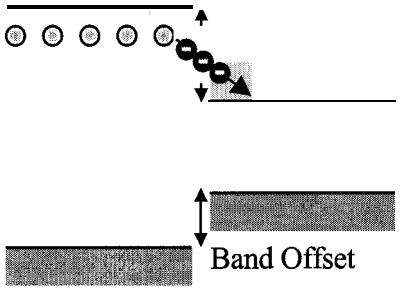


e^-h^+ pair recombines into a photon

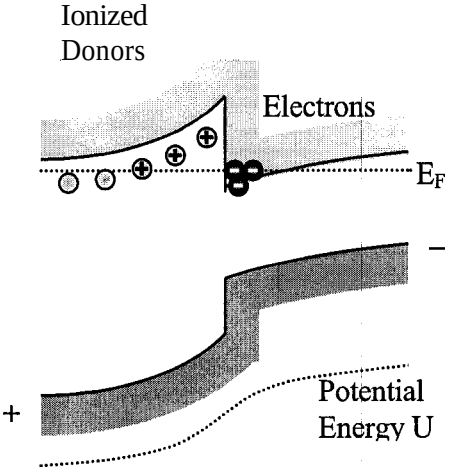
Heteroiunction

Two separate semiconductors:

n-type intrinsic



After forming a junction:



Modulation Doping: Achieve high electron density without scattering of electrons by donors. Record mobilities of $>10^6$ cm^2/Vs. Example: GaAlAs / GaAs

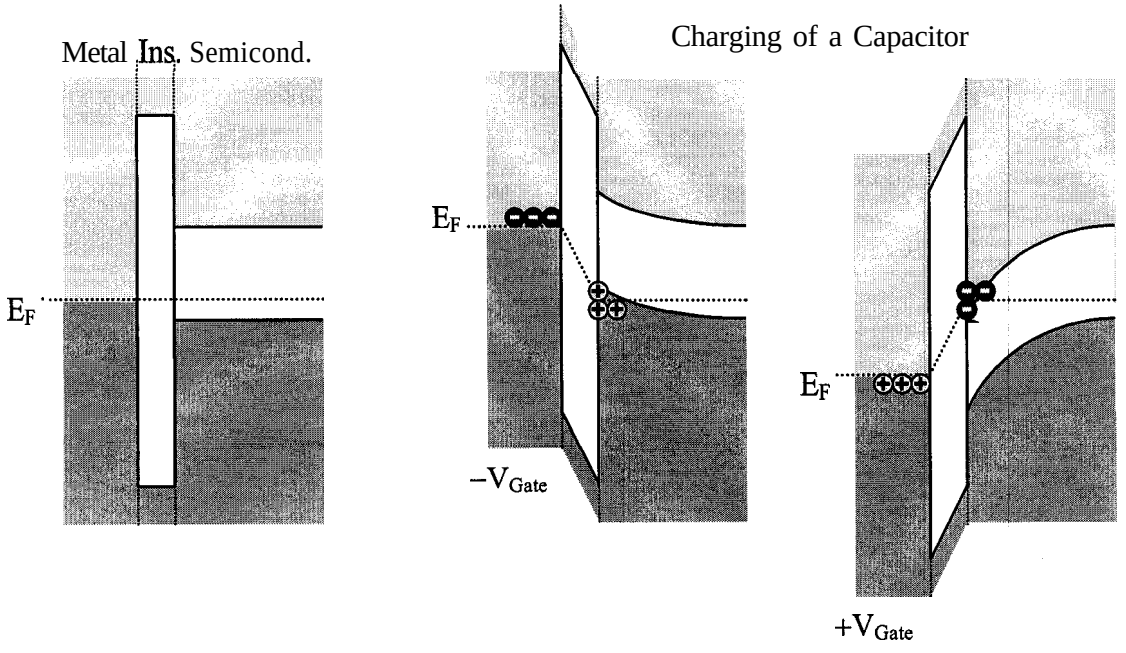
Used for creating a two-dimensional electron gas (quantum Hall effect).

Metal Oxide Semiconductor (MOS)

Shown for a **p-type** semiconductor = NMOS (conduction by **n-type** carriers in inversion)

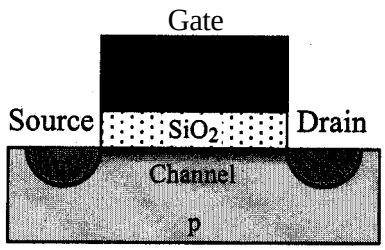
Flatband **Accumulation** **Inversion**

Charging of a Capacitor



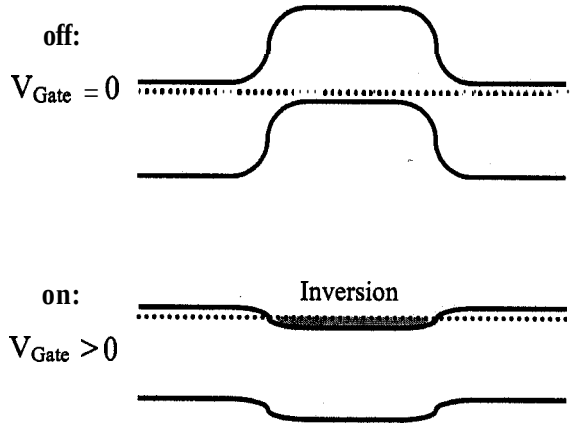
Field Effect Transistor (FET)

NMOS:



Substrate

Channel conductivity controlled by the gate voltage.



Source Channel Drain

= back - to - back Diodes

CMOS (Complementary MOS)

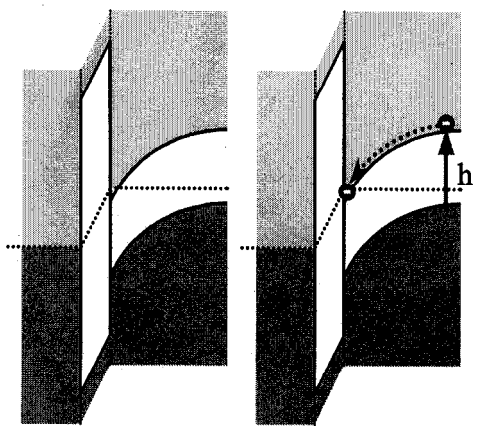
Back-to-back NMOS and PMOS with common gate. Low power consumption, since it draws current during switching only. Most common device in today's electronics.

Quantum Well Laser

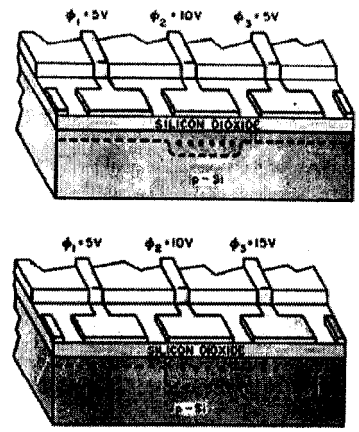
A semiconductor with smaller band gap is inserted at the pn junction of a LED. It traps both electrons and holes and gives them more time to recombine into a photon.

CCD (Charge-Coupled-Device)

Array of MOS capacitors with highly positive gate voltage (inversion). They can store light-induced charge and pass it on along the array for readout.



In the Dark Illuminated



Readout by transfering charge from pixel 2 to pixel 3.